Europe's Energy Reckoning

Post-Russia LNG Dependency, the Renewables Race, and Energy Security

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Executive Summary

Europe's energy architecture has undergone its most consequential restructuring in a generation following the collapse of Russian pipeline gas flows after 2022. The continent has simultaneously pursued three interlocking transformations: a hard pivot from Russian pipeline dependency toward LNG and Norwegian North Sea supply; an accelerated renewables buildout driven by falling solar and wind costs; and a contested attempt to reform carbon pricing and electricity market design to manage the structural contradictions those changes have introduced. None of these transitions is complete, and their interaction — surplus renewable generation coexisting with continued gas dependency for balancing and heat — defines Europe's near-term energy challenge. The ETS, the primary tool for internalising carbon costs, is under political pressure from multiple directions: from industries facing competitiveness stress, and from member states seeking to limit price volatility. How Europe resolves the tension between accelerating decarbonisation and managing industrial exposure will determine whether its energy architecture becomes a durable model or a source of sustained economic fragility.

1. The Post-Ukraine Gas Restructuring: LNG Dependency and the Norwegian Anchor

The 2022 Russian invasion of Ukraine triggered a structural break in European gas supply that has not reversed. Gazprom's pipeline capacity, previously the backbone of continental supply, shifted from a position of relative tightness in 2021 to a situation of very significant spare productive capacity by 2023, as European buyers were cut off or chose to exit long-term contracts. [RAG: LNG_ENERGY_RAG — OIES Oxford Institute for Energy Studies, Natural Gas and LNG Papers, OIES_NG189_RussiaOilGas_2024 p.35]

2. The structural consequence was a reshaping of European import geography. Pipeline flows into Europe stabilised at sharply lower levels through the early 2030s in energy transition scenarios, with further reductions in imports from Russia and North Africa projected, while Power of Siberia 2 redirects Siberian supply toward China rather than westward. [RAG: LNG_ENERGY_RAG — OIES Oxford Institute for Energy Studies, Natural Gas and LNG Papers, OIES_ETScenarios_NatGas_2024 p.24]

3. LNG has filled a significant share of the gap. Globally, LNG exports rose 1.8% to 549 billion cubic metres in 2023, and LNG now accounts for nearly 59% of all globally traded gas, up from a minority share a decade earlier. The United States overtook both Australia and other traditional exporters to become the leading LNG exporter. [RAG: OIL_ENERGY_RAG — BP Energy Outlook 2024 and Energy Institute Statistical Review of World Energy 2024, EI_StatisticalReview2024 p.46]

4. The global emergence of LNG markets has provided structural opportunities to export natural gas that were previously unavailable, the IPCC Sixth Assessment Report notes, driven partly by hydraulic fracturing and directional drilling unlocking unconventional reserves and reducing extraction costs. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

5. In Europe, the net effect on gas demand was nonetheless negative: natural gas consumption dropped in 2022 and had not fully recovered by 2023, when global gas demand rose by only 1 bcm, insufficient to recover 2022 losses. European demand declined further than the global average, reflecting both demand destruction from high prices and a structural shift toward efficiency and electrification. [RAG: OIL_ENERGY_RAG — BP Energy Outlook 2024 and Energy Institute Statistical Review of World Energy 2024, EI_StatisticalReview2024 p.38]

6. Norway has emerged as Europe's indispensable pipeline gas anchor. Norwegian producers have maintained production at even highly mature field complexes by rapidly adding smaller tiebacks and leveraging extensive existing infrastructure. Investment in exploration has expanded, driven explicitly by energy security concerns, including prospective projects in the Barents Sea such as Wisting, though several remain at planning stage. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2024 and 2025, Annual Statistical Bulletin, OPEC_WOO2024 p.162 and OPEC_WOO2025 p.185]

7. The Oxford Institute for Energy Studies maps Norway (NOR) as the dominant supply centre for Northwest European gas pricing, alongside the broader North Sea complex, with pricing linkages running through the TTF hub and into Southeast European markets. The globalisation of the gas market has meant that European spot prices are now materially influenced by Asian demand conditions and global LNG availability — a structural change from the era of bilateral pipeline contracts. [RAG: LNG_ENERGY_RAG — OIES Oxford Institute for Energy Studies, Natural Gas and LNG Papers, OIES_NG195_GasPrices_2024 p.71]

8. Seasonal demand sensitivity remains a material risk. A return to a normal winter, relative to the two mild winters of 2022/23 and 2023/24, can boost European total gas demand by at least 8–10 billion cubic metres; a cold winter adds at least 20–25 bcm. This residual sensitivity, in a market now dependent on global LNG pricing, represents a persistent vulnerability in European energy security planning. [RAG: LNG_ENERGY_RAG — OIES Oxford Institute for Energy Studies, Natural Gas and LNG Papers, OIES_NG202_GlobalGasDemand_2025 p.31]

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2. The ETS Under Strain: Carbon Pricing, Industrial Competitiveness, and Political Pressure

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9. The EU Emissions Trading System, created to place a price on carbon across covered sectors, has delivered measurable emissions reductions: EU carbon output in covered sectors rose 30% since the ETS began in 2005, yet emissions from those sectors have declined — reflecting the counterfactual impact of the scheme. Short-haul airlines such as Ryanair have been drawn into scope as the EU considers extending the carbon border to departing flights. [FT[2026-05-12]]

10. The system nonetheless faces structural stress. The effect of the ETS has been diluted by free allocation provisions for high-emitting industries including steel and chemicals, which reduces the financial incentive to decarbonise. The EU's Carbon Border Adjustment Mechanism (CBAM) was introduced to address carbon leakage risk, but as of mid-2026, the ETS and CBAM combined have failed to create a critical mass of aligned international carbon prices. Economies highly exposed to the EU market — including Turkey, South Korea, and the UK — have introduced their own carbon pricing or harmonised with EU schemes, but only on limited scale. [FT[2026-06-24]]

11. By early 2026, Brussels was actively considering measures to limit carbon prices under the ETS, following pressure from industries facing soaring costs. EU member state support for pausing or capping ETS obligations in the steel sector surfaced in the FT's reporting, reflecting the tension between decarbonisation ambition and industrial competitiveness. [FT[2026-03-26]]

12. The FT's Lex column characterised Europe's CO2 emission allowance as "falling back" and the system as in "need of an overhaul," with pricing having moved in ways that complicate the incentive structure for low-carbon investment. Strong carbon pricing is identified as the underpinning of Europe's decarbonisation framework, but sustaining it in a period of high energy costs and geopolitical stress has proven politically difficult. [FT[2026-02-18]] [FT[2026-07-14]]

13. The IPCC Sixth Assessment Report notes that the EU Market Stability Reserve — a buffer mechanism designed to absorb surplus allowances — has been at least partially successful in stabilising ETS prices in response to short-term disruptions such as the COVID-19 economic shock, providing some evidence that the system's architecture can absorb volatility without collapsing. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

14. The UNEP Emissions Gap Report 2023–24 records that the European Union significantly advanced the Fit for 55 package and REPowerEU in the preceding twelve months, including expansion of the ETS, updates to transport and buildings emissions regulation, and acceleration of the transition away from fossil fuels. These represent the most ambitious legislative packages the EU has enacted, but implementation timelines remain subject to member state capacity and political will. [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023–2024]

15. The Energy Performance of Buildings Directive — a cornerstone of EU demand-side decarbonisation — aims to reduce energy consumption through green home retrofitting across member states. Implementation has been challenging due to inflationary pressures and high living costs, with many countries behind schedule. [RAG: OIL_ENERGY_RAG — OPEC World Oil Outlook 2025, OPEC_WOO2025 p.61]

3. Electricity Market Reform and the Renewables Integration Challenge

16. Europe's electricity market design was built around marginal-cost pricing — where the most expensive generator sets the price for all supply in a given interval. That architecture, functional when fossil fuels dominated, has produced structural anomalies as low-marginal-cost renewables scale up. The FT's coverage of the electricity market reform debate noted proposals from Italy and others to allow member states with more low-carbon sources to decouple from marginal pricing, and illustrated the divergence in outcomes: France's wholesale power market clearing at around €43.43 per MWh compared with €128.14 in Italy. [FT[2026-04-30]]

17. The most visible symptom of the integration problem is negative electricity prices. Europe's negative electricity price events — increasingly common as renewable generation exceeds instantaneous demand — highlight the need for policy reform and robust investment in grid infrastructure. In northern Scotland, wind farms were paid approximately €19 million not to produce power, a curtailment cost reflecting the absence of adequate transmission capacity and storage. [FT[2025-08-21]]

18. Grid-level battery storage is identified as the key enabling technology for resolving the mismatch between generation and demand profiles. Investment in grid-level battery storage is taking off as the market need becomes undeniable. This follows a dramatic reduction in the cost of lithium-ion batteries — down 97% over the past three decades and 90% in the past decade alone — which has fundamentally changed the economics of storage deployment. [FT[2025-08-21]] [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

19. The underlying cost trajectory of renewable generation is strongly supportive of continued buildout. Since 2015, the cost of solar photovoltaics has declined by over 60%; offshore wind costs have fallen by 32% and onshore wind by 23%. Global solar and wind capacities stood at 609 GW and 623 GW respectively in 2019, generating 680 TWh/year and 1,420 TWh/year. Their combined share of global electricity generation reached approximately 8% in 2019, up from around 5% in 2015. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

20. The UK's electricity pricing problem illustrates a broader European tension. High gas prices post-crisis have elevated household electricity bills, yet the structure of the wholesale market — where gas as the marginal generator sets the price even when renewables dominate supply — means consumers do not directly benefit from low-cost renewable generation except through market reform. Wind and solar energy also creates complex and costly network management requirements that add to system costs. [FT[2026-03-02]]

21. Spain's grid experienced a strong oscillation event in which the national operator Red Electrica temporarily cut off the Spanish electricity grid from the rest of the continental system, illustrating the grid stability risks introduced by high renewable penetration without commensurate investment in grid balancing infrastructure. [FT[2025-04-29]]

22. The WMO State of Global Climate 2022–23 confirms that a substantial energy transition is already underway worldwide, with renewable energy generation — driven by solar radiation, wind, and the water cycle — surging to the forefront of climate action. Europe is among the leading regions in deployment density relative to demand. [RAG: CLIMATE_RAG — WMO State of Global Climate 2022–2023]

4. Hydrogen and Long-Horizon Decarbonisation Infrastructure

23. The EU's Projects of Common Interest (PCI) list for hydrogen infrastructure encompasses 65 projects: 31 hydrogen pipeline networks, 10 ammonia import terminals, 17 electrolysers, and 7 hydrogen storage facilities. The majority of hydrogen transmission PCIs and all ammonia import terminal projects are oriented toward import infrastructure, reflecting Europe's strategic intent to source low-carbon hydrogen and ammonia from external producers — North Africa, the Middle East, and potentially hydrogen-exporting regions — rather than relying solely on domestic electrolysis. [RAG: LNG_ENERGY_RAG — OIES Oxford Institute for Energy Studies, Natural Gas and LNG Papers, OIES_NG190_GasToHydrogen_2024 p.82]

24. Pipeline infrastructure repurposing is a central element of the hydrogen strategy. In the UK's Iron Mains Replacement Programme, existing low-pressure gas distribution pipes are being converted from iron to plastic to enable future hydrogen blending or dedicated hydrogen transport. In the Netherlands, an existing low-pressure 12 km natural gas pipeline has been converted to hydrogen use, providing a proven proof-of-concept for the broader network. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

25. Hydrogen-based energy carriers — including ammonia and synthetic hydrocarbons (Liquid Organic Hydrogen Carriers, or LOHCs) — carry an advantage in that they can be transported using existing oil tankers and storage tanks, significantly reducing the infrastructure investment required relative to direct hydrogen shipping. LOHC projects transporting hydrogen using existing oil infrastructure are already under development, pointing toward a near-term commercialisation pathway. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

26. Energy security considerations explicitly link hydrogen to the broader European supply diversification strategy. The IPCC AR6 notes that energy security is perceived as a national security issue and is frequently prioritised over climate concerns in policymaking — a dynamic that in Europe's post-Ukraine context has reinforced rather than undermined the case for hydrogen, since domestic or allied-source

green hydrogen reduces geopolitical energy exposure. [RAG: CLIMATE_RAG — IPCC Sixth Assessment Report (AR6) 2021–2023]

Outlook

Europe enters the second half of the 2020s with its energy architecture materially more resilient than in 2021, but still structurally exposed. The Russian gas dependency has been broken at high economic cost; LNG has provided a functional replacement, but at spot-market pricing that introduces persistent volatility and winter-demand sensitivity that pipeline contracts previously dampened. Norway has become the indispensable physical anchor for pipeline supply, with investment expanding under energy security logic — but production at mature fields is subject to plateau dynamics, and the Barents Sea upside remains speculative.

The carbon pricing architecture faces its most serious political test. The ETS was designed to tighten over time and drive capital allocation toward low-carbon investment. That mechanism is working in the sense that emissions in covered sectors have fallen, but the political cost — competitive stress on steel, chemicals, and energy-intensive industry — is generating pressure for price caps, free allocation extensions, and ETS pauses. If that pressure succeeds in decoupling carbon prices from the trajectory required to meet Fit for 55 targets, the EU's industrial decarbonisation pathway will extend significantly, with consequences for climate commitments and for the competitiveness logic of CBAM.

The electricity sector is at the most advanced stage of transition, and also the most structurally complex. The scaling of renewables has outpaced grid infrastructure and market design reform. Negative prices, curtailment payments, and grid oscillation events signal that the next phase of investment must prioritise transmission, interconnection, and storage over generation addition alone. Electricity market reform — shifting away from pure marginal-cost pricing toward mechanisms that better reflect the long-run cost structure of a low-carbon system — is the defining regulatory challenge for European energy in the near term.

Hydrogen infrastructure development is on a longer timeline. The EU PCI pipeline reflects genuine policy commitment, but the majority of project financing remains to be secured, and the economics of green hydrogen remain sensitive to electrolyser cost trajectories and renewable electricity prices. Ammonia import terminals offer a nearer-term pathway, leveraging existing maritime and storage infrastructure, and are likely to be the first hydrogen-carrier imports to reach commercial scale.

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